The Sun

THE SUN IS OUR NEAREST STAR. ITS HEAT AND LIGHT MAKE ALL LIFE ON EARTH POSSIBLE.

Although 100 times wider than Earth, the Sun is an average star in terms of its size and age. Studying the Sun has helped us understand how other stars in the Universe work.

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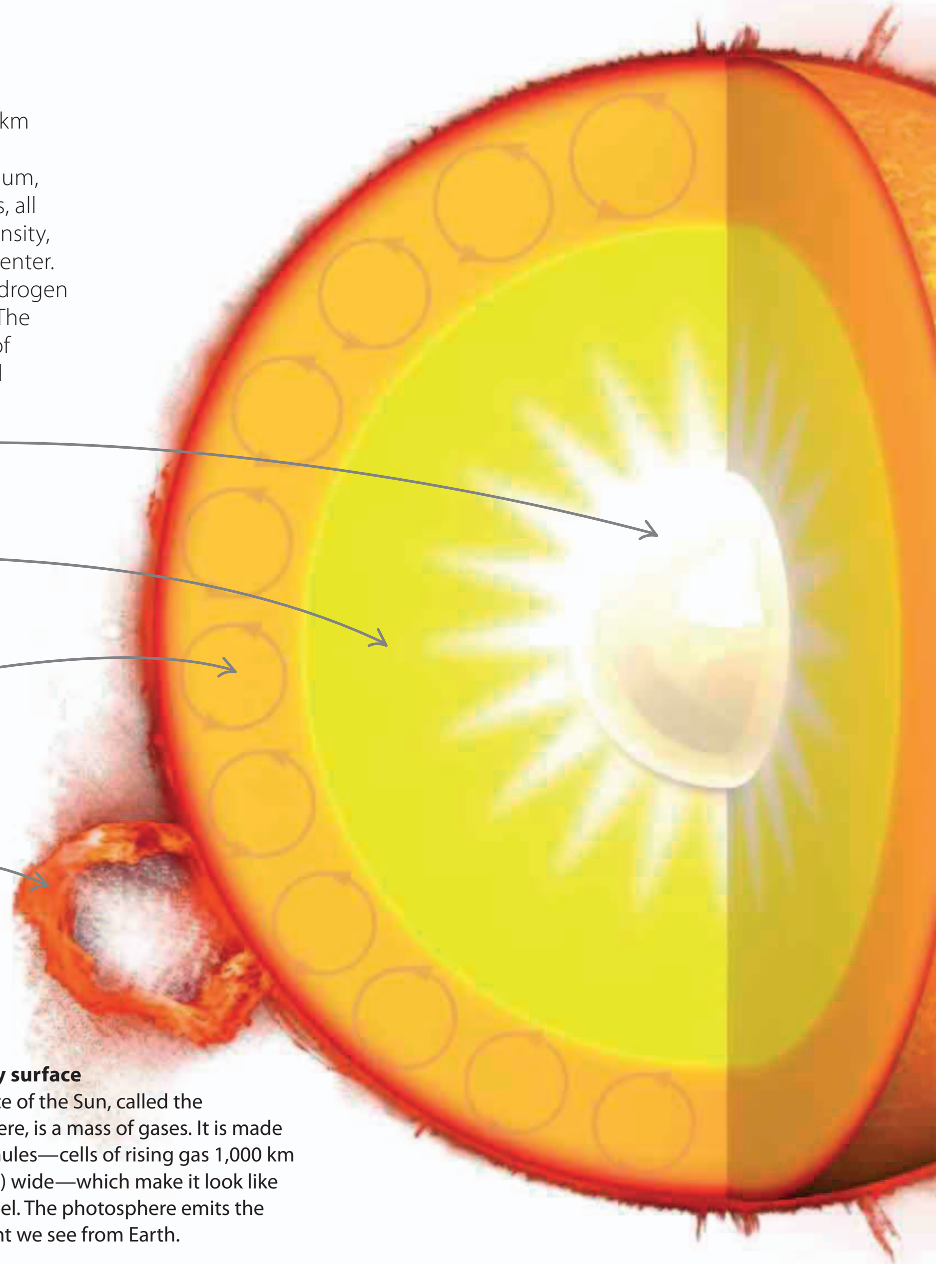
Inside the Sun

The Sun is an immense ball of gas 1.4 million km (870,000 miles) wide. It is made up of almost three-quarters hydrogen, about a quarter helium, and small amounts of 65 or so other elements, all held together by gravity. The temperature, density, and pressure of the gas increase toward the center. At the core, nuclear reactions that convert hydrogen to helium are the source of the Sun's energy. The energy radiates out, taking many thousands of years to reach the surface, where it is released into space as light and heat.



▷ **Stormy surface**
The surface of the Sun, called the photosphere, is a mass of gases. It is made up of granules—cells of rising gas 1,000 km (620 miles) wide—which make it look like orange peel. The photosphere emits the visible light we see from Earth.

The Sun's mass is about **750 times greater** than all the other objects in the Solar System put together.